Toward a Trustworthy Classifier With Deep CNN: Uncertainty Estimation Meets Hyperspectral Image

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***Abstract*— Recently, deep convolutional neural networks (CNNs) have achieved a high classification accuracy of hyperspectral images (HSIs). However, high accuracy is not the only goal of a good HSI classifier. In real-world applications, it is necessary to tell whether the classifier is certain about its classification result, which is critical for safe usage. Unfortunately, most of the existing models do not consider the issue. In this study, uncertainty is estimated and reduced to build a trustworthy HSI classifier. First, since the output probabilities of the softmax layer cannot represent the confidence scores, the distance measurement scheme is used to measure the confidence scores. Then, a trustworthy HSI classifier, which reduces the predictive uncertainty in CNN (i.e., PU-CNN), is obtained by minimizing the distance to the correct centroid. Second, the fact that a training sample of HSI usually contains many pixel vectors that belong to different classes brings label uncertainty. Then, label uncertainty CNN (i.e., LU-CNN), which uses a classifier-consistent estimator to recover the multiple classes in each HSI sample, is proposed. LU-CNN computes loss over candidate label sets to find the optimal classes, which leads to a trustworthy HSI classifier. Finally, the combination of PU-CNN and LU-CNN [i.e., predictive uncertainty and label uncertainty (PL-CNN)] is proposed to address predictive uncertainty and label uncertainty at the same time. Experimental results on the three popular hyperspectral datasets show that the proposed methods yield improvements in both accuracy and confidence. The proposed trustworthy classifier opens a new window for the safe usage of HSI.**

***Index Terms*— Confidence, convolutional neural network (CNN), hyperspectral image (HSI) classification, uncertainty estimation.**

I. Introduction

HYPERSPECTRAL imaging sensor is able to provide rich spectral information with hundreds of spectral bands, which could distinguish the subtle differences of diverse materials of interest [1]. The obtained hyperspectral images (HSIs) have been successfully applied to many areas, including agriculture [2], military [3], and forestry [4]. Among HSI processing techniques, classification is one of the most important topics, which tries to allocate each pixel to a predefined category (i.e., road, water, and so on) [5].

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Conventional HSI classification approaches are mainly focused on handcrafted feature extraction including spectral features [6], spatial features [7], or spectral–spatial features [8]. Representative feature extraction works included the principal component analysis (PCA) [9], Gabor filtering [10], morphological attribute profiles [11], and so on. Then, these extracted features were fed into classical classifiers, including the K-nearest neighbors [12] and support vector machines (SVMs) [13], to finish the HSI classification task.

Nevertheless, the above HSI classification methods were not in an end-to-end manner. Recently, the deep learning-based approaches achieved a significant improvement in many research areas [14]. Inspired by the success of deep learning models, many researchers designed a series of new and improved convolutional neural network (CNN)-based classifiers for HSI classification, which can be roughly divided into four types: 1-D-CNN [15], 2-D-CNN [16], 3-D-CNN [17], and CNN-related approaches [18].

Due to the fact that HSI has abundant spectral information, 1-D-CNN extracted the spectral features with a 1-D convolutional kernel. For example, in [19], a novel classification framework based on pixel-pair features was obtained by 1-D-CNN for HSI classification. In addition, considering the high dimensionality of the spectral domain, Yu *et al* utilized 1-D-CNN to reduce the spectral dimension of HSI, which mitigated the overfitting problem and achieved better classification performance [20]. Expect for learning discriminant spectral features with a single 1-D convolutional kernel, spectral–contextual details were explored to learn the position invariant features by few 1-D convolutional layers in [21].

Because spatial information was also important for HSI classification, 2-D-CNN was explored to learn the spatial features [22]. For example, in [23], 2-D CNNs were leveraged to improve HSI classification by automatically learning the data-adaptive kernels. Furthermore, to extract spatial features with different shapes, deformable sampling locations were investigated in [24]. Besides, some handcrafted features (i.e., Gabor features) were combined with 2-D-CNN to increase the textures information, which alleviated the overfitting problem with limited training samples [25]. In addition, due to the fact that only limited training samples were available for HSI, the adaptive spatial pyramid constraint method was proposed to address this problem [26].

In addition, due to the fact that HSI was a cube, extensive studies directly handled the HSI with 3-D convolutional kernels for spectral–spatial feature extraction [27]. In [28],

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a multiscale 3-D deep CNN was investigated for HSI classification, which could jointly learn both multiscale spatial and spectral features. Moreover, to add ingrained texture information, a generalized gradient centralized 3-D convolution operation was proposed to extract both the intensity level and gradient level information [29].

Expect for the above classical convolutional architectures, some works related to CNN were developed for HSI classification in recent years [30]. Specifically, the graph convolutional network (GCN) is an emerging network architecture, which has advantages in modeling relations between samples. For example, Hong *et al*. proposed a miniGCN strategy to reduce computational cost and improve classification performance for HSI classification [31]. Except for the single GCN, a dual GCN including the point graph network and the distribution graph network was also proposed for the supervised classification of HSI [32]. However, GCN is difficult to handle the relationships of sequential spectral data in HSI, which is important for better classification performance. To address this problem, Transformer was introduced based on the self-attention mechanism to learn long-range dependencies of HSI in combination with CNN [33]. Besides, in [34], spectrally local sequence information from neighboring bands of HSI was learned by Transformer to yield groupwise spectral embeddings. However, sufficient labeled samples are usually required for deep CNN-based methods to obtain satisfactory accuracy. To address the problem of limited training samples in HSI classification, CNN-based self-supervised learning and weakly supervised learning were proposed [35]. For instance, self-supervised learning with an adaptive distillation strategy was proposed to address the problem of insufficient training samples in HSI [36].

In spite of the fact that comprehensive studies have obtained high overall accuracy (OA) for HSI classification, few works considered trustworthiness as the goal for HSI classification. Trustworthiness is crucial to precision agriculture, such as vegetation classification [37]. There would be a great concern if the CNN-based HSI approaches are implemented in practical applications. For example, a classifier needs to refuse the low confidence results. Therefore, only using the accuracy as an evaluation metric is not enough, and the trustworthiness is also important for a CNN-based HSI classifier.

To establish a trustworthy classifier, uncertainty in the classification system should be measured and reduced. In general, uncertainty in a CNN-based HSI classifier arises from two different sources: model uncertainty and data uncertainty [38]. Model uncertainty refers to the uncertainty, which is caused by the processing of designing and training the models. In addition, data uncertainty occurs because of the overlapping samples and systematic label noise.

Several methods focused on uncertainty have been proposed for HSI unmixing, restoration, and denoising. Specifically, in HSI unmixing, the endmember uncertainty was related to whether the HSI pixels fit into the endmember simplex. The proposed method estimated the model uncertainty of the endmembers directly from the linear mixing model [39]. Besides, in HSI restoration, to select relevant bands of HSI, a symmetric uncertainty coefficient was proposed based on mutual information [40]. In addition, in HSI denoising, uncertainty quantification was proposed to address the difficulty of handling data uncertainty in HSI patch mixing [41].

Although uncertainty has been investigated in HSI unmixing, restoration, and denoising, to the best of our knowledge, no related studies have been found. Due to the fact that low uncertainty leads to an HSI trustworthy classifier, two types of uncertainty, i.e., predictive uncertainty and label uncertainty, are explored in this study. Specifically, the predictive uncertainty estimates the uncertainty of predictions of CNN. Most of the CNN-based methods choose the highest softmax probability as a confidence score. However, for the samples that are classified wrongly, the max values of the probabilities are still high [42]. To address this issue, we replace the softmax layer in CNN with a novel distance measurement method. During the training, the predictive uncertainty in CNN (PU-CNN) is reduced by minimizing the distance, and the minimum distance represents the confidence scores.

Another uncertainty is the label uncertainty, for HSI classification, considering the process of selecting samples for spectral–spatial information extraction, data uncertainty on the label is investigated, which indicates that many pixel vectors corresponding to different classes mismatch a label included in each HSI spectral–spatial sample. To estimate and reduce label uncertainty CNN (LU-CNN), we design a classifier-consistent estimation for the classifier, which could infer multilabel for each training sample.

This article explores a general framework for establishing a trustworthy CNN-based HSI classifier, which is a good start toward raising awareness of the importance of uncertainty in HSI classification. We analyze the predictive uncertainty and label uncertainty, which would influence the output of the classifier for HSI classification. Then, a unified framework for estimating predictive uncertainty and label uncertainty CNN (PL-CNN) simultaneously is established. All the proposed methods achieve competitive classification accuracy and enhance the trustworthiness of CNN for HSI classification. The main contributions of this article are listed as follows.

1) To the best of our knowledge, it is the first time that the concept of the trustworthy classifier is proposed for the HSI classification task, which improves the classification accuracy and reduces the uncertainty of the classifier at the same time. Furthermore, the expected calibration error (ECE) is introduced to evaluate the confident level of the HSI classification methods.
2) PU-CNN, which reduces the predictive uncertainty by decreasing the distance between the CNN's output and all centroids, is proposed for HSI trustworthy classification. Distinct from CNN, PU-CNN obtains an excellent confidence score while maintaining competitive accuracy.
3) LU-CNN, which reduces the label uncertainty by minimizing a novel classifier-consistent estimator, is explored for HSI trustworthy classification. Unlike the classical CNN-based HSI classifier, LU-CNN improves the label confidence in each spectral–spatial sample for HSI classification.

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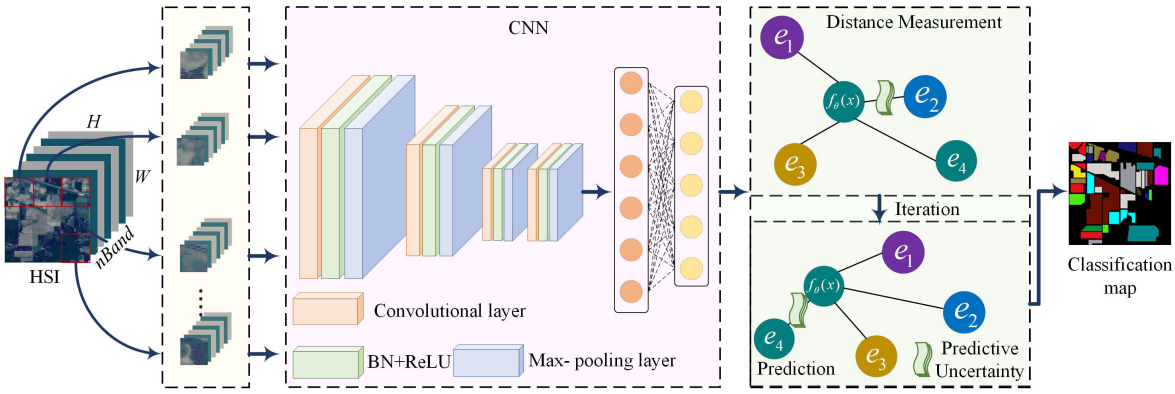


Fig. 1. Overall schematic framework of PU-CNN.

4) PL-CNN, which combines PU-CNN and LU-CNN to establish a CNN-based HSI classifier, is proposed to decrease the predictive uncertainty and label uncertainty simultaneously. The proposed PL-CNN achieves better accuracy and confident score compared to PU-CNN and LU-CNN.

The rest of this article consists of three sections. Section II introduces the proposed methods, including PU-CNN, LU-CNN, and PL-CNN in detail. Section III presents experimental settings, analyses, results, and discussions for HSI classification. Finally, the conclusion of this study is drawn in Section IV.

II. Methodology

In this section, three approaches (i.e., PU-CNN, LU-CNN, and PL-CNN) are proposed to reduce the predictive uncertainty or/and label uncertainty for trustworthy HSI classifiers.

A. Preliminary

To better understand the context of this article, several definitions are elaborated as follows.

1) The predictive uncertainty is caused by the output probabilities of the softmax layer, which cannot stand for the confidence scores in CNN for HSI classification.

2) The label uncertainty results from the fact that many pixel vectors corresponding to different classes mismatch a label included in each HSI spectral–spatial sample.

3) *Data Preparation Process:* Let $I' \in R^{H\times W\times B}$ be an HSI; H, W, and B represent the height, the width, and the number of spectral bands of HSI, respectively. I' is divided into $H\times W$ samples for data preparation. Each pixel i in I' generates a fixed square box with the size of $S\times S$. Therefore, a sample set $I = \{I_1, \ldots, I_{HW}\}, I_i \in R^{S\times S\times B}$ is constructed. The corresponding label set is defined as $Y = \{y_1, \ldots, y_{HW}\}, y_c \in \{1, \ldots, C\}$, where C indicates the total number of classes in the HSI.

B. Predictive Uncertainty CNN (PU-CNN) for HSI Classification

For an HSI classification task, CNN takes the maximum value of the predicted probabilities, which are given by the softmax layer output, as the prediction. However, the probabilities of the network cannot indicate how confident a model is.

To solve this issue, the proposed PU-CNN estimates and reduces the predictive uncertainty to make the prediction probabilities reflect the confidence scores. The framework of the proposed PU-CNN is shown in Fig. 1. There are two main parts in PU-CNN, including a classical CNN and a predictive uncertainty-based processing mechanism. Detailed descriptions on these parts are presented in the following.

First, the processed HSI samples are fed into a classical CNN to learn hierarchical special–spatial features. The CNN consists of convolutional layers, max-pooling layers, activation functions (i.e., ReLU), and a fully connected layer. The key building layer in a classical CNN is the convolutional layer, which can be defined as

$$O^u = \sum W_a^u * v_a^{u-1} + b \tag{1}$$

where O^u indicates the output feature map at the uth layer. v_a^{u-1} is the ath feature map at the $(u-1)$th layer. W is the weights in convolutional filters. $*$ indicates the convolution operator, and b represents the bias.

Next, a predictive uncertainty-based processing mechanism replaces the softmax layer with the distance measurement scheme, and the minimum distance represents the confidence scores.

Specifically, the centroid of class C (i.e., e_c) is initialized as normal distribution with mean zero and standard variance 0.05. e_c is updated by

$$e_C = (W_C \cdot y_C)/N_{C,t} \tag{2}$$

where y_C indicates the ground truth of class C and the length of e_c is n. W_C is the class dependent weight matrix with the size of $n\times d$, and d represents the output dimension of CNN. $N_{C,t}$ is computed in (6).

K_c represents the distance between the output of CNN without the softmax layer and e_C, and it is computed as follows:

$$K_C(f_\theta(I_i), e_C) = \exp[-\frac{\frac{1}{n}||W_C f_\theta(I_i) - e_C||_2^2}{2\sigma^2}] \tag{3}$$

where f_θ indicates the trained CNN. σ is the length scale, which is a hyperparameter.

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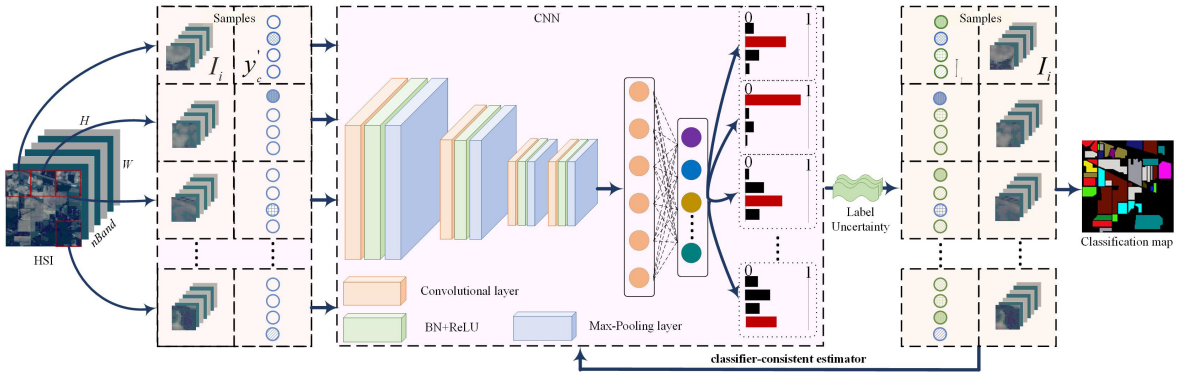


Fig. 2. Overall schematic framework of LU-CNN.

The prediction of each sample I_i with label y_C aims at looking for the maximum correlation between I_i and a set $E = \{e_1, \ldots, e_C\}$, which contains centroids of different classes. The update of E uses an exponential moving average of the feature vectors of I_i belonging to the corresponding y_C. Then, a prediction p_i is made by taking the class C with the minimum distance between input HSI sample and e_C, and the calculated minimum distance indicates the confidence score of I_i

$$p_i = \arg\min K_C(f_\theta(I_i), e_C). \tag{4}$$

During the training process, we minimize the distance to the correct centroid while maximizing the distance to the wrong centroid. The corresponding loss function is constructed as follows:

$$l_{PU} = -\sum_C y_C \log(K_C) + (1 - y_C)\log(1 - K_C) \tag{5}$$

where $K_C(\cdot, e_C)$ represents the kernel value of per class and K_C is the abbreviation of $K_C(f_\theta(x), e_C)$.

However, the above updating rule may suffer from the closed form solution, which leads to the centroids with θ and W held constant. To solve this problem, exponential moving averages [43] is updated by

$$N_{C,t} = \gamma \times N_{C,t-1} + (1 - \gamma) \times n_{C,t} \tag{6}$$

$$m_{C,t} = \gamma \times m_{C,t-1} + (1 - \gamma) \times \sum_i W_c f_\theta(I_{C,t,i}) \tag{7}$$

$$e_{C,t} = \frac{m_{C,t}}{N_{C,t}} \tag{8}$$

where $n_{C,t}$ indicates to the total number of I_i assigned to the same class c in minibatch t, $I_{C,t,i}$ refers to the element i of a minibatch at time t with class C, and γ represents the momentum. $m_{C,t}$ represents the exponential moving average of an HSI sample, and $N_{C,t}$ is exponential moving averages of all the HSI samples.

However, there exists the feature collapse problem, which indicates that the features of samples are far away from a particular right centroid. To avoid this problem, the proposed PU-CNN enforces the CNN change to the various input, which identifies different classes distinctively.

Therefore, we define the following formula to enforce the model to be sensitive to the input samples, which regularizes the feature representations by using a two-sided gradient penalty, $\lambda \cdot [||\nabla_x \sum_C K_C||_2^2 - 1]^2$, where λ represents the gradient penalty weight.

By decreasing the distances between the CNN's output and centroids of different classes, the minimum distances represent the confidence scores of different HSI samples. The high confidence scores are obtained in the correct predictions, while the incorrect predictions have low confidence scores.

C. Label Uncertainty CNN (LU-CNN) for HSI Classification

For most of the deep learning-based methods, each HSI spectral–spatial training sample is annotated with a single label. However, many pixel vectors corresponding to different classes are included in each HSI sample, which brings label uncertainty.

In order to solve this problem, in this article, LU-CNN is proposed to estimate and reduce the label uncertainty, which infers multiple labels in each training sample by designing a classifier-consistent estimator, such that the CNN-based HSI classifier is confident about the input.

The overall pipeline of the proposed LU-CNN is shown in Fig. 2. Here, y_c' with a blue circle mark refers to the one-hot label of each training sample I_i. By designing a novel classifier-consistent estimator, the proposed LU-CNN could recover multiple labels concealed in each training sample I_i, which are displayed in green circle marks.

Specially, instead of using a single label y_c of I_i, a candidate label set $s \in S, S = \{P(Y_c)/\phi/Y_c\}$ is defined to estimate the label uncertainty in each training sample, where S is full of labeling information without the empty and whole label information. For simplicity, suppose that an HSI dataset has three different land-cover classes. If the given ground truth is attributed to the second class, the corresponding label is represented by $\{2\}$. S is the power set of the label space $\{1, 2, 3\}$ except for the empty set and the whole label set, i.e., $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3\}$. In the proposed LU-CNN, a classifier is trained with access only to the labels, which includes the second class, such as $\{2\}$, $\{1, 2\}$, and $\{2, 3\}$ to find the optimal candidate label set, for example, $\{1, 3\}$. Therefore, the formula is defined as follows:

$$P_{(I,Y)\sim p(I_i,Y_c),S\sim P(s|I_i,Y_c)}(Y_c \in s) = 1 \tag{9}$$

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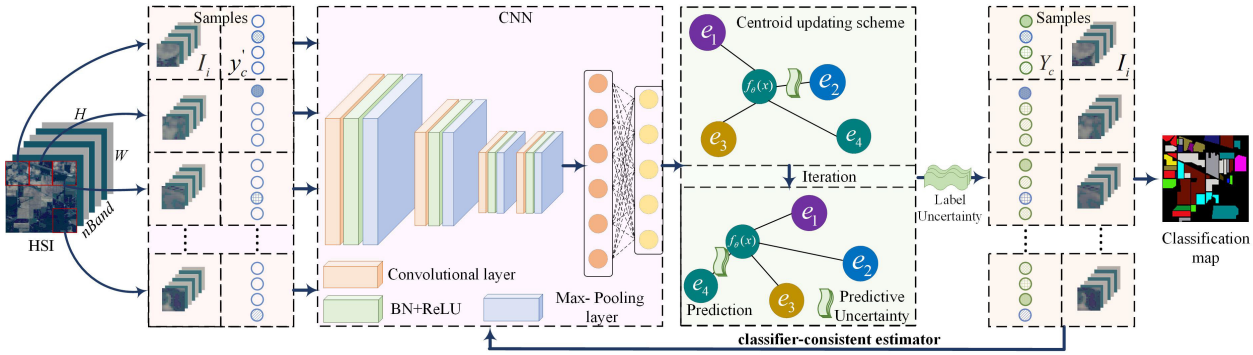


Fig. 3. Overall schematic framework of PL-CNN.

where Y_c represents the estimated label of I_i, $Y_c \in$ s. $p(s|I_i, Y_c)$ is the class-conditional density.

The formula of the risk estimator aims to induce the classifier to assign multiple labels in each HSI sample

$$R(g) = E(I, S) \sim p(I_i, s)[l_{LU}(g(I), S)] \tag{10}$$

where l is the loss of LU-CNN and g represents an HSI classifier.

To compatible with efficient optimization of the CNN, an estimation error bound is established in this article; (10) is modified as follows to make the loss estimable by minimizing loss over the candidate label set s:

$$l_{LU}(g(I), S) = \min_{i \in S} l_{LU}(g(I), y_c') \tag{11}$$

where y_c' indicates the one-hot label and a new risk estimator is given by

$$R(g) = E_{(I,S)\sim p(I_i,s)} \min_{i \in S} l_{LU}(g(I), y_c'). \tag{12}$$

Consider that (12) is difficult to optimize with the *min* operator since the wrong label y_c' is always wrong from the beginning to the end.

Therefore, (11) is modified by relaxing the min operator with dynamic weights; instead of l being computed on the all training set, we compute l on every label

$$l_{LU}(g(I), y_c') = \sum_{i=1}^{c} l_{LU}(g_i(I), y_{ci}') \tag{13}$$

where y_{ci}' represents the ith element of y_c'.

Therefore, the empirical risk estimator is transformed by

$$\widehat{R}(g) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{c} w_{ij} l_{LU}(g_j(I_i), y_{cj}'^{s_i}) \tag{14}$$

where $y_{cj}'^{s_i}$ indicates the jth coordinate of $y_c'^{si}$; $y_{cj}'^{s_i} = \sum_{k \in s_i} y_c'^{k}$. s_i is the ith candidate label set. w_{ij} is the label weight jth of w_i and $w_i \in \Delta^{c-1}$, and the confidence of the jth label is one of the multiple labels of I_i. Here, $w_{ij} = 1$ when $\arg\min_{k \in s_i} l_{LU}(g(I_i), y_c'^{k}) = j$ and 0 otherwise.

Eventually, j is one of the multiple labels of I_i, which is estimated with $w_{ij} = 1$. Inspired by the expectation maximization (EM) algorithm [44], the uniform weights are used to initialize the model

$$w_{ij} = \begin{cases} 1/|s_i|, \text{if } j \in s_i, \\ 0, \quad \text{otherwise.} \end{cases} \tag{15}$$

However, in the few epochs of the CNN, the "frequent patterns" is remembered [45], [46]. Thus, a classifier would be guided by the correct labels, which gives relatively low losses for correct labels. To tackle this problem, the weights are estimated by using the current predictions for slightly putting more weights on more possible labels

$$w_{ij} = \begin{cases} g_j(I_i) / \sum_{k \in s_i} g_k(I_i), \text{if } j \in s_i, \\ 0, \text{otherwise.} \end{cases} \tag{16}$$

In this fashion, $\widehat{R}(g)$ is the final classifier-consistent estimator. When the minimal $\widehat{R}(g)$ is obtained, multiple labels in each spectral–spatial training sample are progressively identified in w_{ij}.

D. Predictive Uncertainty and Label Uncertainty CNN (PL-CNN) for HSI Classification

To establish a trustworthy CNN-based HSI classifier, two kinds of uncertainties, i.e., predictive uncertainty and label uncertainty, are investigated at the same time in the proposed PL-CNN for HSI classification.

Fig. 3 displays the flowchart of the proposed PL-CNN. Specifically, each HSI sample and a single label are given to the CNN. During the training process, instead of choosing the max value of outputs in the softmax layer, we minimize the distance between the CNN's output and the predefined centroids corresponding to the different classes. The prediction of each sample is assigned to the closest centroid. Then, these predictions update the classifier-consistent estimator by numerous iterations, which reduces the label uncertainty. Finally, multiple labels in each spectral–spatial sample are identified.

The total loss function is a summation of l_{PU} in PU-CNN and l_{LU} in LU-CNN defined above; β_1 and β_2 are set to balance their importance

$$l_{PL} = \beta_1 l_{PU} + \beta_2 l_{LU}. \tag{17}$$

Therefore, given the input HSI samples, the predictive uncertainty and label uncertainty are estimated and reduced by optimizing the loss simultaneously.

E. Expected Calibration Error

In order to evaluate the confident level of the proposed methods (i.e., PU-CNN, LU-CNN, and PL-CNN),

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the ECE is adopted to measure the trustworthiness of HSI classification [47].

Specifically, the first step is to group predictions into M interval bins. Thus, each bin has a size of $1/M$. Then, the accuracy of each bin is calculated. Here, let B_m be the set of indices of samples, in which predictions attribute to the interval $I_m = (m - 1/M, m/M]$. $acc(B_m)$ can be calculated as follows:

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} 1(\widehat{y}_i = y_i) \tag{18}$$

where $\widehat{y}_i$ indicates the predicted value of sample i and y_i is the ground truth of sample i. Next, the average confidence within B_m is described as follows:

$$conf(B_m) = \frac{1}{|B_{\text{m}}|} \sum_{i \in B_m} \widehat{p}_i \tag{19}$$

where $\widehat{p}_i$ represents the predicted value of sample i.

The target of a trustworthy classifier should be

$$acc(B_m) = conf(B_m). \tag{20}$$

However, the above method lacks in estimating the number of samples; from a statistical point of view, ECE computes the difference in expectation between the confidence and accuracy, and trustworthiness is orthogonal to accuracy

$$E[P(\widehat{Y} = Y|\widehat{P} = p) - p]. \tag{21}$$

The final ECE is given by

$$ECE = \sum_{m=1}^{M} \frac{|B_m|}{n} ||acc(B_m) - conf(B_m)|| \tag{22}$$

where n indicates the total number of samples.

In each bin, the difference between accuracy and confidence indicates a certain gap. The small gap leads to a trustworthy classifier for HSI classification.

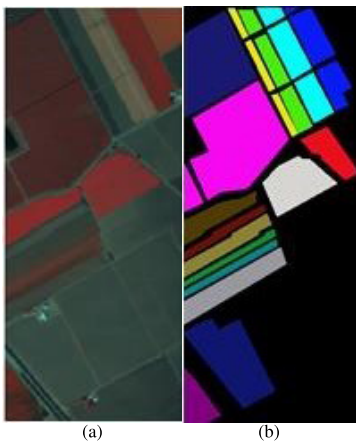
III. Results

A. Data Description

The proposed methods (i.e., PU-CNN, LU-CNN, and PL-CNN) are evaluated on the three public datasets, including the Salinas, Pavia, and Indian Pines datasets; the details about these datasets are described as follows.

1) Salinas: This dataset was captured by the airborne visible imaging spectrometer (AVIRIS) sensor over Salinas Valley, California, which consists of 204 bands after 20 water absorption bands (108–112, 154–167, and 224) are removed. This dataset was composed of 512 × 217 pixels, whose spatial resolution is 3.7 m. There are 16 different land-cover classes. The corresponding false composite image and the number of samples in each class are shown in Fig. 4 and Table I, respectively.

2) Pavia: The Pavia dataset was collected by the Reflective Optics System Imaging Spectrometer (ROSIS-3) over the Pavia University. It is composed of 610 × 340 pixels, and the spatial resolution is 1.3 m. 103 bands were used for analysis in the range of 0.43–0.86 μm. The ground reference map covers nine classes of interest, and the detailed information is shown in Fig. 5 and Table II, respectively.



(a) (b)

Fig. 4. Salinas dataset. (a) False-color composite image. (b) Ground-truth map.

TABLE I
SALINAS LABELED SAMPLE COUNTS

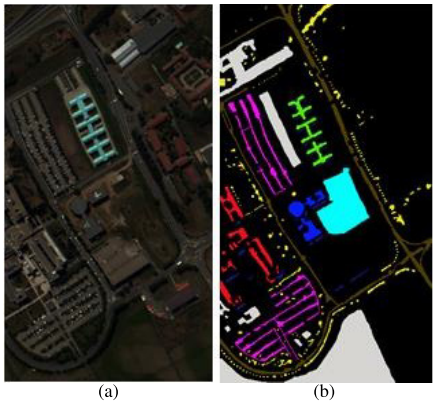
Class			Sample Numbers
No.	Color	Name	
1		Brocoli-green-weeds-1	1977
2		Brocoli-green-weeds-2	3726
3		Fallow	1976
4		Fallow-rough-plow	1394
5		Fallow-smooth	2678
6		Stubble	3959
7		Celery	3579
8		Grapes-untrained	11213
9		Soil-vinyard-develop	6197
10		Corn-senesced-green-weeds	3249
11		Lettuce-romaine-4wk	1058
12		Lettuce-romaine-5wk	1908
13		Lettuce-romaine-6wk	909
14		Lettuce-romaine-7wk	1061
15		Vinyard-untrained	7164
16		Vinyard-vertical-trellis	1737
Total			53785

3) Indian Pines: The available dataset was gathered by the AVIRIS sensor over the Indian Pines test data with 220 bands. There are 200 bands left ranging from 0.4 to 2.5 μm, after removing 20 water absorption and low signal-to-noise ratio bands (104–108, 150–163, and 220). It is composed of 145 × 145 pixels with a 20-m spatial resolution. 16 classes are provided in this scene; the reference map and the number of samples in each class are displayed in Fig. 6 and Table III, respectively.

B. Experimental Design

In the experiments, each dataset is split into three subsets, including the training set, the validate set, and the test set. Specifically, the training dataset consists of 200 samples, which are randomly selected from all the labeled samples; the validate set includes 50 samples chosen from the leaving samples; and the test set consists of all the remained samples. Here, the validate set is used to evaluate the performance of the model during the training process.

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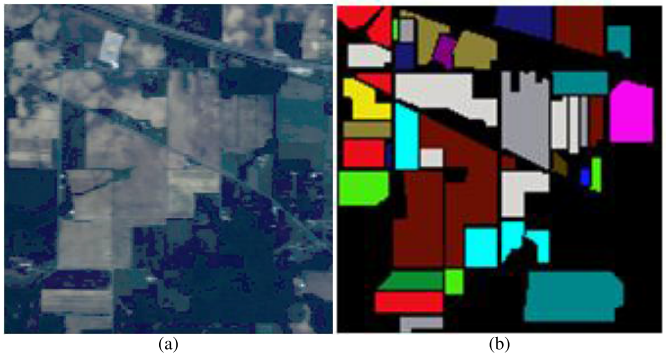


(a) (b)

Fig. 5. Pavia dataset. (a) False-color composite image. (b) Ground-truth map.

TABLE II
PAVIA LABELED SAMPLE COUNTS

Class			Sample
No.	Color	Name	Numbers
1		Asphalt	6851
2		Meadows	18686
3		Gravel	2207
4		Trees	3436
5		Metal sheets	1378
6		Bare soil	5104
7		Bitumen	1356
8		Bricks	3878
9		Shadow	1026
Total			43922



(a) (b)

Fig. 6. Indian Pines dataset. (a) False-color composite image. (b) Ground-truth map.

In the experiments, in order to fully utilize the spectral–spatial information of the HSI, each labeled pixel with its 33×33 neighborhood is selected as a sample during the preprocessing step. Four classical approaches (i.e., extended morphological profile (EMP)-SVM, CNN, spectral–spatial residual network (SSRN), and HybridSN) are adopted for comparison to purposefully illustrate the validity of the proposed methods. Specifically, the compared method EMP-SVM consists of two important parameters named C and γ, and a set of $\{10^{-3}, 10^{-2}, \ldots, 10^{3}\}$ is designed to find the optimal values of both parameters. The spatial information of HSI is extracted by the morphological opening and closing operations, which adopts the disk-shaped structuring element with a size of two, four, six, and eight in EMP-SVM. For a fair comparison, the hyperparameter settings of CNN, SSRN, and HybridSN are the same as [25], [48], and [49], respectively.

TABLE III
INDIAN PINES LABELED SAMPLE COUNTS

Class			Sample
No.	Color	Name	Numbers
1		Alfalfa	1977
2		Corn-notill	3726
3		Corn-min	1976
4		Corn	1394
5		Grass-pasture	2678
6		Grass-trees	3959
7		Grass-pasture-mowed	3579
8		Hay-windrowed	11213
9		Oats	6197
10		Soybean-notill	3249
11		Soybean-mintill	1058
12		Soybean-clean	1908
13		Wheat	909
14		Woods	1061
15		Buildings-Grass-Trees	7164
16		Stone-Steel-Towers	1737
Total			53785

In addition, a classical approach named Monte-Carlo (MC)-Dropout is utilized as a comparison for HSI classification [50]. The MC-Dropout has been proven a single and effective method with dropping out extracted information from existing models that have been thrown away so far.

All the methods are evaluated on the three datasets with five quantitative metrics, including the OA, average accuracy (AA), kappa coefficient (Kappa), the accuracy of each class, and the ECE for measuring the confidence level of different methods.

C. Parameter Analysis

To verify the influence of parameters of all the proposed methods, parameters are analyzed on the three datasets with 200 training samples. For the proposed PU-CNN and PL-CNN, there are two key parameters. The first one is the length scale σ, which affects the distance between the output of the model and the centroids. The second one is the weight for gradient penalty λ, which influences the sensitivity of the model. The grid search method is utilized to find both optimal parameters σ and λ, which vary from 0.1 to 1.

The related experimental results are shown in Figs. 7 and 8. For the proposed PU-CNN, it can be observed that the OA is relatively stable when σ and λ change on the Indian Pines dataset. However, the larger values of the parameter σ achieve better performance than the small values on the Salinas dataset. According to the results of OA, the values of σ and λ are set to 0.9 and 0.8 on the Salinas dataset; 0.2 and 0.2 on the Pavia dataset; and 1 and 0.8 on the Indian Pines dataset, respectively. For the proposed PL-CNN, the big values ofσ and λ would result in a good performance on the Pavia dataset. Besides, OA follows a parabolic trend when the value of λ is about 0.2, and accuracy reaches an optimal point on the Salinas dataset.

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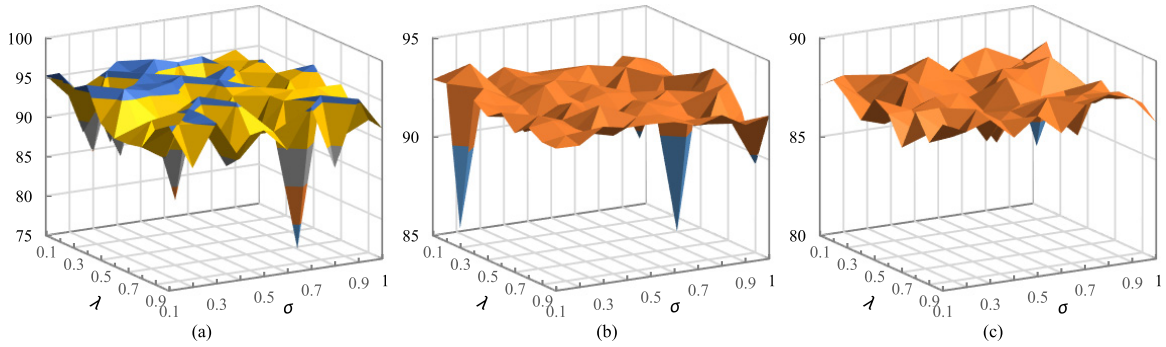


Fig. 7. Influence of the length scale σ and gradient penalty λ of PU-CNN on the three datasets. (a) Salinas. (b) Pavia. (c) Indian Pines.

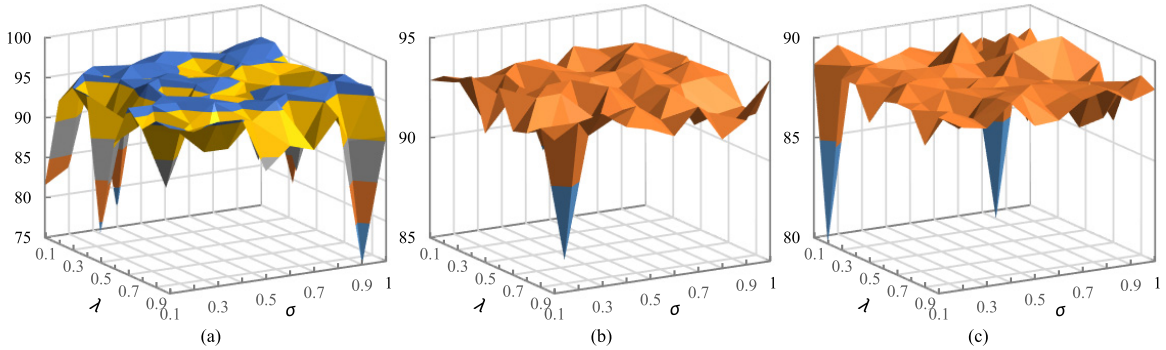


Fig. 8. Influence of the length scale σ and gradient penalty λ of PL-CNN on the three datasets. (a) Salinas. (b) Pavia. (c) Indian Pines.

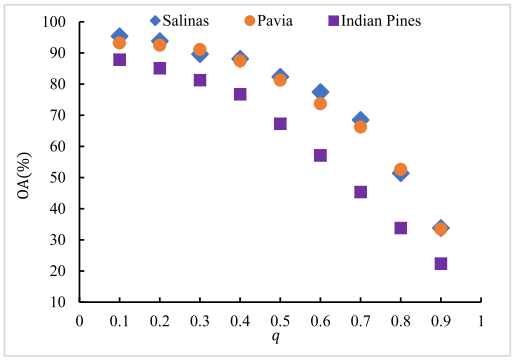


Fig. 9. Results of LU-CNN with different values of q on the three datasets.

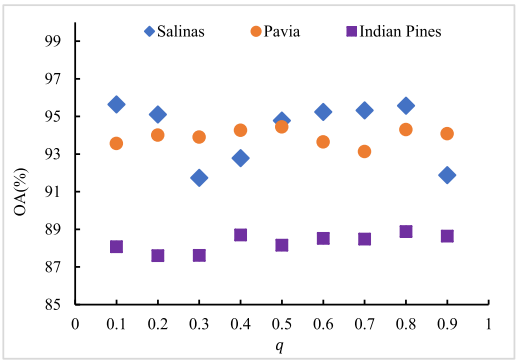


Fig. 10. Results of PL-CNN with different values of q on the three datasets.

Based on the experimental results, the optimal σ and λ are set to 0.2 and 1, 1 and 1, and 0.2 and 0.4 on the Salinas, Pavia, and Indian Pines datasets, respectively.

A key parameter flipping probability q in the proposed LU-CNN and PL-CNN is to initialize each training label in a candidate label set. The value of q is from 0.1 to 1with intervals of 0.1 to search for the optimal value. Fig. 9 shows the results of LU-CNN with different values of q. It can be observed that OA decreases as the value of q grows. The larger the values of q is, the more difficult it is to search for labels of each HSI sample, resulting in poorer performance. Therefore, the value of q is set to 0.1 on all the datasets to identify multiple labels of each HSI sample in the experiments. In addition, Fig. 10 displays the influence of parameter q different values; the classification accuracy is affected by the parameter q to varying degrees. It can be found that the highest OA of q is obtained with 0.1, 0.5, and 0.8 on the Salinas, Pavia, and Indian Pines datasets, respectively. Therefore, we set the above values to achieve better performance for HSI classification.

D. Experimental Setting and Analysis of the Proposed PU-CNN

PU-CNN estimates and reduces the predictive uncertainty of CNN by establishing a distance measurement scheme for HSI classification, which outputs the confidence scores of the CNN-based HSI classifier in a single forward pass. During the

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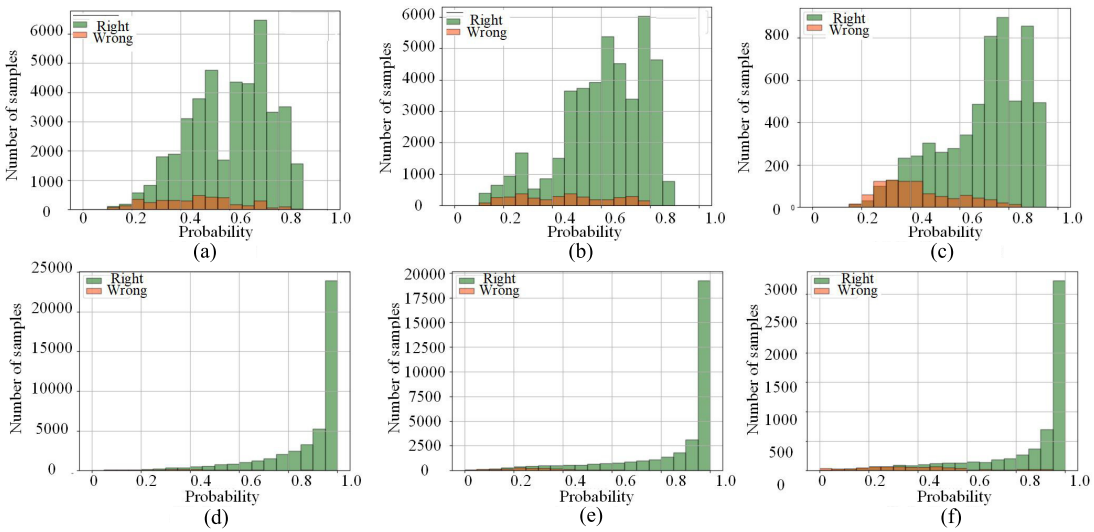


Fig. 11. Histogram of the probability of EMP-SVM and the proposed PU-CNN on the three datasets. (a) Salinas. (b) Pavia. (c) Indian Pines. (d) Salinas. (e) Pavia. (f) Indian Pines.

training process, the learning rate is set to 0.2. The total number of epochs is set to 300, and the minibatch is adopted with the value of 400.

To show that the proposed PU-CNN could increase the trustworthiness of the classifier by reducing predictive uncertainty; Fig. 11 displays the distribution of confidence scores (i.e., probabilities associated with the predicted label) of the classical compared method EMP-SVM and the proposed PU-CNN on the three datasets. The horizontal axis represents the confidence score of each sample, and the vertical axis indicates the distribution of test samples with respect to output probabilities. The green bar and the orange bar denote the number of correctly and incorrectly classified test samples in each interval, respectively.

More specifically, Fig. 11(a)–(c) shows the confidence scores for each sample in EMP-SVM, which takes the largest probabilities as the confidence scores. Fig. 11(d)–(f) displays the confidence scores in the proposed PU-CNN on the Salinas, Pavia, and Indian Pines datasets, respectively. Take the Pavia dataset as an example [see Fig. 11(b)]; the orange bars are spread out over the middle part, while, for Fig. 11(e), the orange bar locates at the left part. From these figures, it can be seen that confidence scores of the proposed PU-CNN of the correct predictions are high while for the incorrect predictions are low.

E. Experimental Setting and Analysis of the Proposed LU-CNN

The proposed LU-CNN could recover the ground-truth label concealed in each HSI training sample by reducing the label uncertainty in the CNN-based HSI classifier. During the training procedure, the number of epochs, the learning rate, and the batch size is set to 200, 0.001, and 400 for LU-CNN on all the datasets, respectively.

To illustrate the effectiveness of the proposed LU-CNN, one compared classical method CNN is chosen; Fig. 12 visualizes the confidence score of the proposed LU-CNN [see Fig. 12(a)–(c)] and the compared method CNN [see Fig. 12(d)–(f)] on the Salinas, Pavia, and Indian Pines, respectively. It can be observed that the number of the samples is most classified correctly; in addition, the original bar locates at the left part, while the green bar is distributed at the right part. It demonstrates that LU-CNN works better with label uncertainty than CNN.

F. Experimental Setting and Analysis of the Proposed PL-CNN

PL-CNN aims to reduce the predictive uncertainty and label uncertainty at the same time. In the experiments of the PL-CNN, the validated set is used to determine the hyperparameters on all the datasets; the number of training epochs, the batch size, and the learning rate is empirically set to 300, 400, and 0.2, respectively.

In the proposed PL-CNN, β_1 and β_2 are the two adjustable parameters in the loss function with a range of (0, 1]. To verify the classification effect of the PL-CNN in the case of different values of β_1 and β_2, some experiments have been done. Fig. 13 shows the OA on the three datasets when the parameters β_1 and β_2 take different values. It can be seen that both parameters are sensitive to the classification performance. In the experiments, to obtain the highest OA, the appropriate values of β_1 and β_2 are set to 1 and 0.5 on the Salinas dataset and 1 and 0.2 on the Pavia and Indian datasets, respectively.

To illustrate the proposed method PL-CNN could also increase the confidence level of the CNN-based HSI classifier; Fig. 14 displays the histogram of the confidence scores on the three datasets. Among them, (a)–(c) and (d)–(f) show the confidence scores in the compared method MC-Dropout and the proposed PL-CNN, respectively. From these figures, it can be seen that the proposed PL-CNN is able to separate the test samples classified correctly and incorrectly. Specifically, most of the test samples classified correctly are close to one on the three datasets; test samples that are classified incorrectly distribute at the left part of the horizontal axis. The results

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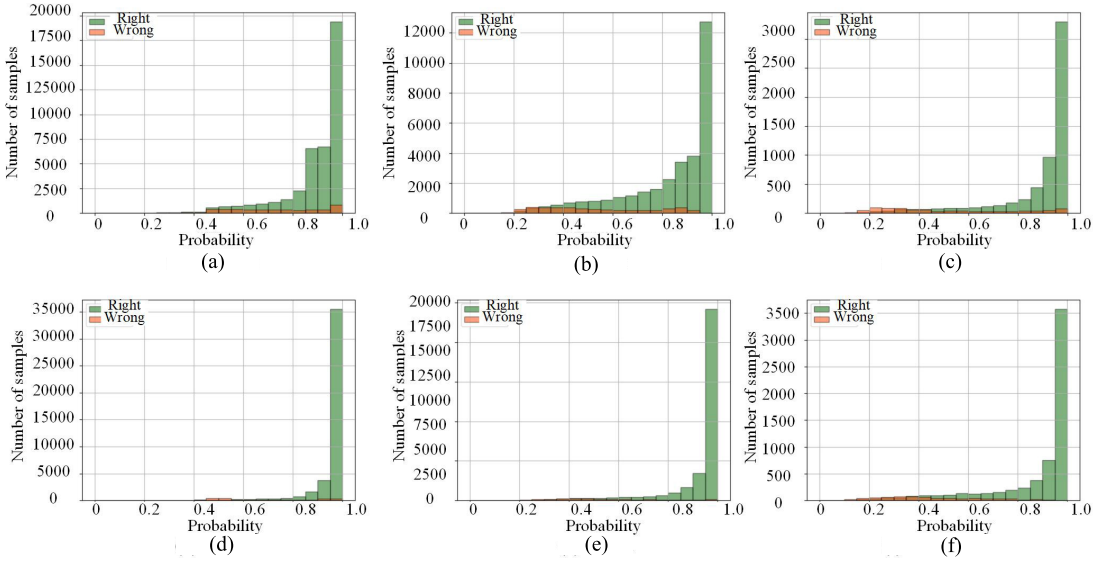


Fig. 12. Distribution of confidence scores of CNN and the proposed LU-CNN on the three datasets. (a) Salinas. (b) Pavia. (c) Indian Pines. (d) Salinas. (e) Pavia. (f) Indian Pines.

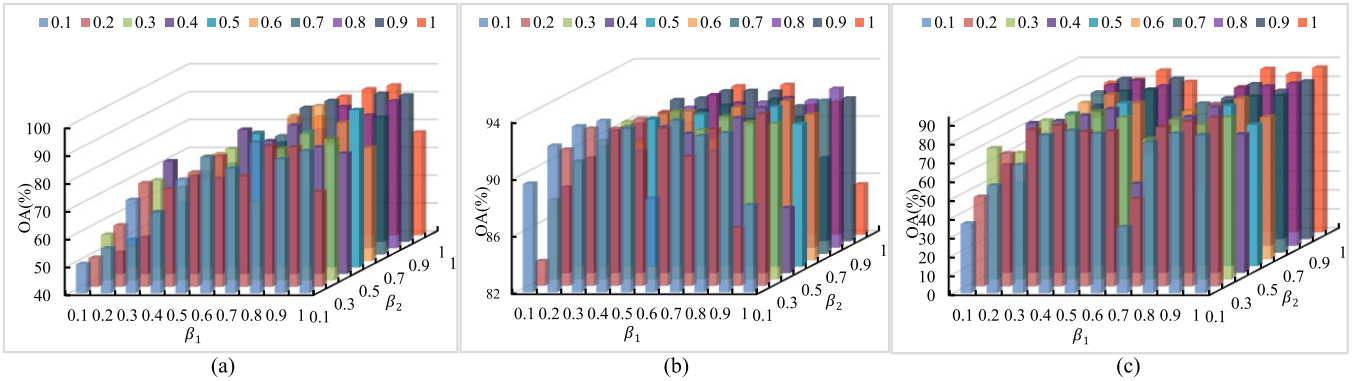


Fig. 13. Influence of the length scale σ and gradient penalty λ of PU-CNN on the three datasets. (a) Salinas. (b) Pavia. (c) Indian Pines.

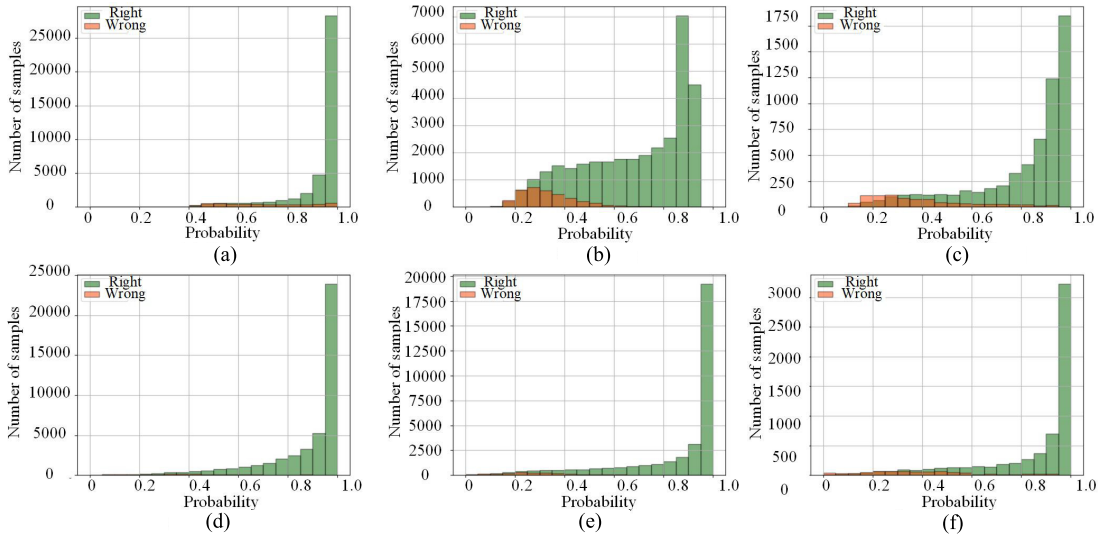


Fig. 14. Distribution of confidence scores of MC-Dropout and the proposed PL-CNN on the three datasets. (a) Salinas. (b) Pavia. (c) Indian Pines. (d) Salinas. (e) Pavia. (f) Indian Pines.

demonstrate that the correct predictions have high confidence scores, while, for the incorrect predictions, the confidence scores are low.

G. Ablation Study

To validate the effectiveness of each component (i.e., model uncertainty and label uncertainty) in the proposed methods

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TABLE IV
ABLATION EXPERIMENTS FOR MODEL UNCERTAINTY AND DATA UNCERTAINTY ON THE SALINAS DATASET

CNN	Model uncertainty	Data uncertainty	OA (%)	AA (%)	K×100	ECE (%)
✓			89.79 ±8.30	90.10 ±4.25	88.70±9.02	13.61±6.72
✓	✓		95.82±1.51	94.22±2.93	95.31±1.70	3.32±2.12
✓		✓	95.38±2.16	92.02±4.30	94.81±2.43	3.50±1.62
✓	✓	✓	96.14±0.83	93.79±2.16	95.65±0.92	1.61±0.78

TABLE V
ABLATION EXPERIMENTS FOR MODEL UNCERTAINTY AND DATA UNCERTAINTY ON THE PAVIA DATASET

CNN	Model uncertainty	Data uncertainty	OA (%)	AA (%)	K×100	ECE (%)
✓			91.62±0.59	82.95±1.82	89.43±0.71	9.06±2.98
✓	✓		93.84±1.23	87.94±1.87	92.20±1.59	4.26±1.24
✓		✓	93.20±1.43	85.69±3.35	91.42±1.81	4.00±1.62
✓	✓	✓	94.08±0.70	87.38±1.35	92.51±0.87	1.35±0.65

TABLE VI
ABLATION EXPERIMENTS FOR MODEL UNCERTAINTY AND DATA UNCERTAINTY ON THE INDIAN PINES DATASET

CNN	Model uncertainty	Data uncertainty	OA (%)	AA (%)	K×100	ECE (%)
✓			86.16±2.12	65.79±6.58	83.35±2.55	9.06±2.98
✓	✓		88.68±1.19	72.94±7.34	86.46±1.44	7.60±1.95
✓		✓	87.84±1.72	67.88±5.26	85.43±2.08	4.22±1.59
✓	✓	✓	89.00±2.03	73.80±7.05	86.82±2.46	2.18±0.53

(i.e., PU-CNN, LU-CNN, and PL-CNN), we perform an ablation study on the three HSI datasets. The mean OA, AA, K, and ECE are shown in Tables IV–VI on the Salinas, Pavia, and Indian Pines datasets, respectively. From these results, it can be found that CNN with model uncertainty or/and label uncertainty improves the classification accuracy by a large margin compared with CNN on all the datasets. Besides, adding both components (i.e., model uncertainty and label uncertainty) to CNN is better than one component (i.e., model uncertainty or label uncertainty) in terms of OA, AA, and K on the three datasets. In terms of ECE, compared to CNN, the value of ECE of CNN with model uncertainty and label uncertainty is the lowest, which is below 5% on all the datasets. Therefore, model uncertainty and label uncertainty in the proposed methods are both effective components for HSI classification. They work collaboratively to render satisfactory classification performance and confidence level of the CNN-based HSI classifier.

H. Classification Results of All the Proposed Methods

The detailed classification results of PU-CNN on the three datasets with 200 training samples are presented in the third to last columns in Tables VII–IX. From the results, one can see that the proposed PU-CNN exhibits the highest accuracies on all the datasets. EMP-SVM obtains low classification results compared to the other deep learning-based methods with discriminate spectral–spatial feature extraction. The proposed PU-CNN outperforms all other deep learning-based methods (i.e., CNN, SSRN, MC-Dropout, and HybridSN); taking the Salinas dataset as an example, PU-CNN increases the OA with CNN, SSRN, MC-Dropout, and HybridSN by 6.03%, 5.01%, 4.74%, and 4.56%, respectively. The above results verify the efficiency of the proposed PU-CNN.

The results of the LU-CNN are exhibited in the second last column. As shown in Tables VII–IX, in comparison to other methods, LU-CNN gains the highest accuracy on all the datasets. For example, compared to the MC-Dropout, LU-CNN improves the OA by 4.3%, 2.23%, and 1.71% on the Salinas, Pavia, and Indian Pines datasets, respectively. In addition, in terms of OA on the Salinas dataset, LU-CNN achieves improvement by 7.79%, 5.59%, 4.57%, and 4.12% in comparison to EMP-SVM, CNN, SSRN, and HybridSN, respectively. The above phenomena show that the proposed LU-CNN increases the accuracies of the three datasets.

For PL-CNN, it can be found that PL-CNN improves the classification accuracy by a large margin at the last column in Tables VII–IX. The OA of the PL-CNN reaches 96.14%, 94.08%, and 89.00% on the Salinas, Pavia, and Indian Pines

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TABLE VII

CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE SALINAS DATASET USING 200 TRAINING SAMPLES

Method	EMP-SVM	CNN	SSRN	MC-Dropout	HybridSN	**PU-CNN**	**LU-CNN**	**PL-CNN**
OA(%)	87.59±2.39	89.79 ±8.30	90.81 ±2.87	91.08±4.76	91.26±3.17	95.82±1.51	95.38±2.16	**96.14±0.83**
AA(%)	88.65±1.93	90.10 ±4.25	90.77 ±3.27	92.15±3.76	90.46±3.96	**94.22±2.93**	92.02±4.30	93.79±2.16
K×100	86.15±2.75	88.70±9.02	89.89 ±3.25	90.01±5.34	90.27±3.53	95.31±1.70	94.81±2.43	**95.65±0.92**
ECE (%)	31.00±3.00	13.61±6.72	21.59±5.64	12.13±2.98	5.96±1.86	3.32±2.12	3.50±1.62	**1.61±0.78**
Brocoli-green-weeds-1	95.28±4.21	66.96±38.74	47.59±63.61	96.31±7.18	**99.53±0.68**	98.23±2.80	91.09±18.28	96.66±3.72
Brocoli-green-weeds-2	98.49±0.28	99.47±0.85	**99.99±0.02**	96.88±5.87	99.88±0.21	98.99±1.08	98.25±2.65	99.09±0.83
Fallow	70.53±22.58	96.05±4.79	65.27±9.61	90.27±18.71	87.06±16.91	95.64±7.17	88.65±18.74	**97.61±2.60**
Fallow-rough-plow	**99.80±0.12**	98.06±2.40	99.19±0.63	98.88±1.31	86.68±14.06	96.66±2.90	97.48±5.40	97.33±2.79
Fallow-smooth	93.34±4.40	**97.94±2.14**	94.16±8.27	97.94±2.43	92.29±9.29	97.09±3.31	93.55±8.41	95.61±4.44
Stubble	99.36±0.22	99.91±0.11	99.10±0.98	**99.98±0.03**	98.26±2.48	99.87±0.27	98.80±2.88	99.97±0.05
Celery	97.48±1.84	99.59±0.42	98.87±1.56	**99.68±0.48**	97.66±2.52	99.24±0.77	99.06±1.11	99.58±0.71
Grapes-untrained	90.09±7.83	72.24±37.12	83.95±9.71	79.91±19.12	88.21±4.01	93.91±1.47	95.74±2.71	**95.91±2.67**
Soil-vinyard-develop	98.89±0.08	99.85±0.18	**99.98±0.04**	99.54±0.58	99.73±0.56	99.62±0.46	99.79±0.37	99.62±0.46
Corn-senesced-green-weeds	90.74±2.23	93.54±8.37	93.96±1.35	95.65±4.92	85.54±12.82	91.68±8.78	**98.20±2.87**	95.08±4.84
Lettuce-romaine-4wk	93.06±2.02	90.14±11.32	90.53±12.29	**93.52±6.89**	78.36±11.74	87.39±12.65	86.66±20.77	92.02±8.14
Lettuce-romaine-5wk	**99.98±0.04**	95.51±5.28	92.17±10.92	97.40±2.65	90.06±8.95	98.65±1.51	85.05±31.86	97.64±3.02
Lettuce-romaine-6wk	77.91±39.03	99.18±0.98	**100±0.00**	98.65±1.55	79.08±16.66	96.38±7.94	80.24±22.50	95.44±9.36
Lettuce-romaine-7wk	98.34±0.99	98.81±1.76	91.85±9.60	99.25±1.50	91.44±8.47	**99.76±0.31**	99.02±1.69	99.16±1.87
Vinyard-untrained	35.73±30.40	89.74±12.88	83.68±4.81	78.71±38.01	83.29±7.32	91.71±9.75	**94.61±6.32**	89.92±9.48
Vinyard-vertical-trellis	79.35±6.22	44.59±30.50	**91.59±10.84**	51.82±32.59	90.23±11.12	58.26±25.36	66.14±35.18	49.94±35.83

TABLE VIII

CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE PAVIA DATASET USING 200 TRAINING SAMPLES

Method	EMP-SVM	CNN	SSRN	MC-Dropout	HybridSN	**PU-CNN**	**LU-CNN**	**PL-CNN**
OA(%)	89.43±1.20	91.62±0.59	92.00±2.01	90.97±1.81	92.18±1.35	93.84±1.23	93.20±1.43	**94.08±0.70**
AA(%)	80.37±3.60	82.95±1.82	**87.88±4.61**	80.57±3.38	82.45±1.82	87.94±1.87	85.69±3.35	87.38±1.35
K×100	85.81±1.64	89.43±0.71	89.52±2.72	88.55±2.35	89.60±1.81	92.20±1.59	91.42±1.81	**92.51±0.87**
ECE (%)	15.00±2.00	9.06±2.98	23.23±5.62	7.76±2.45	5.90±1.40	4.26±1.24	4.00±1.62	**1.35±0.65**
Asphalt	91.84±1.75	91.64±4.03	94.99±3.82	**95.70±1.04**	95.47±3.65	95.3±2.08	93.47±2.54	95.24±2.49
Meadows	98.02±0.68	99.17±0.33	97.62±1.87	99.16±1.24	**99.74±1.36**	99.80±0.38	99.04±1.20	99.28±0.49
Gravel	66.63±14.10	46.16±18.32	61.45±12.11	45.16±25.57	**91.94±9.07**	76.14±5.99	43.83±21.09	66.08±12.41
Trees	92.10±5.14	90.66±4.27	**96.34±2.68**	86.90±10.70	88.95±6.13	89.91±7.30	96.03±1.44	92.80±2.57
Metal sheets	86.26±21.92	97.58±3.33	99.49±0.26	98.35±1.52	**100±0.00**	98.75±1.23	95.74±3.98	97.44±1.83
Bare soil	78.66±7.12	99.25±0.69	81.84±15.59	96.91±2.42	**99.98±6.90**	89.77±10.74	99.31±1.23	96.82±2.60
Bitumen	**74.43±10.12**	63.31±18.21	70.61±39.42	56.03±26.97	70.44±5.83	62.23±17.81	60.72±14.78	65.46±11.28
Bricks	90.92±3.70	85.97±7.89	91.06±1.70	85.32±10.31	62.85±13.24	93.45±3.00	92.15±5.01	**95.17±1.32**
Shadow	74.11±22.21	72.84±16.96	97.49±1.80	61.62±18.65	32.71±14.04	80.93±12.53	**90.89±6.73**	78.09±10.29

TABLE IX

CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE INDIAN PINES DATASET USING 200 TRAINING SAMPLES

Method	EMP-SVM	CNN	SSRN	MC-Dropout	HybridSN	**PU-CNN**	**LU-CNN**	**PL-CNN**
OA(%)	81.53±2.39	86.16±2.12	82.37±6.45	86.13±1.15	83.77±4.58	88.68±1.19	87.84±1.72	**89.00±2.03**
AA(%)	72.43±6.35	65.79±6.58	67.44±7.68	66.04±5.73	68.19±12.95	72.94±7.34	67.88±5.26	**73.80±7.05**
K×100	78.87±2.78	83.35±2.55	79.87±7.32	83.31±1.42	80.95±5.49	86.46±1.44	85.43±2.08	**86.82±2.46**
ECE (%)	54.15±21.03	9.06±2.98	23.22±1.32	7.76±2.45	7.69±3.89	7.60±1.95	4.22±1.59	**2.18±0.53**
Alfalfa	50.27±41.37	50.82±32.72	16.60±36.48	47.14±29.41	30.67±41.11	54.38±33.34	30.25±30.01	**62.14±36.81**
Corn-notill	72.43±7.51	85.52±7.43	71.55±22.58	88.82±5.91	85.63±8.08	89.18±3.15	**90.04±3.51**	88.09±4.73
Corn-min	**80.77±5.42**	68.18±9.96	79.21±22.83	67.86±9.83	69.16±16.26	70.85±11.05	57.98±24.61	73.48±10.98
Corn	**74.44±19.28**	43.26±32.56	48.43±28.18	44.25±33.39	52.04±32.70	59.73±37.82	50.60±35.44	55.80±39.14
Grass-pasture	77.04±8.10	60.95±12.54	**84.05±11.08**	62.82±9.26	76.98±19.39	71.51±15.24	64.99±17.47	65.05±15.44
Grass-trees	95.27±3.54	91.54±6.50	94.17±8.41	91.53±9.76	**97.41±2.04**	93.51±7.16	94.86±2.20	95.62±4.21
Grass-pasture-mowed	49.63±49.64	42.61±24.59	26.28±42.86	46.61±31.47	36.98±50.64	55.91±37.37	45.33±38.56	**57.14±37.99**
Hay-windrowed	**97.43±2.90**	84.14±19.09	96.25±5.91	87.15±11.50	87.73±24.64	93.88±7.54	80.79±34.62	93.25±10.65
Oats	28.95±39.68	20.00±40.00	3.51±8.59	20.00±40.00	**49.47±46.72**	20.00±44.72	36.47±28.85	20.00±44.72
Soybean-notill	74.92±7.72	89.02±6.31	82.35±15.78	87.69±7.32	84.90±7.36	86.81±4.14	**92.24±2.88**	86.16±5.76
Soybean-mintill	85.30±4.85	95.86±2.49	84.04±11.11	94.34±2.48	88.43±4.58	95.12±2.05	93.41±2.52	**96.08±1.37**
Soybean-clean	60.49±10.01	48.14±20.02	75.61±19.87	44.24±12.99	51.43±23.82	70.25±21.94	66.08±22.36	**76.61±15.49**
Wheat	75.54±37.88	77.07±38.85	76.11±37.70	77.67±38.89	79.07±20.48	76.95±43.20	**92.83±9.50**	73.64±41.98
Woods	96.56±3.24	98.57±0.94	**99.17±1.06**	98.28±1.35	95.62±3.64	98.59±1.18	98.43±2.00	97.96±2.08
Buildings-Grass-Trees	73.77±9.55	51.22±34.29	**74.95±17.30**	47.75±34.26	60.13±25.38	61.85±42.47	44.88±25.28	65.91±40.35
Stone-Steel-Towers	66.09±43.03	45.78±22.17	66.79±45.21	50.46±20.01	45.32±38.23	68.50±31.58	46.87±29.94	**73.88±24.69**

datasets, respectively, which outperforms other comparative models. More specifically, taking the more complex Pavia dataset as an example, PL-CNN is higher than MC-Dropout by 3.11%, 6.81%, and 3.96% in terms of OA, AA, and K. Besides, the proposed PL-CNN outperforms CNN, SSRN, and HybridSN by 2.46%, 2.08%, and 1.9% in terms of OA on

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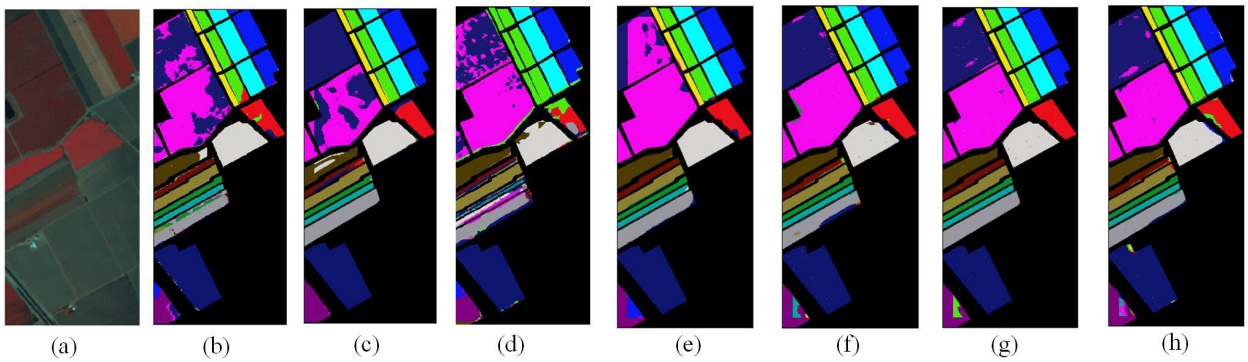


Fig. 15. Salinas. (a) False-color composite image. The classification maps using (b) EMP-SVM, (c) CNN, (d) SSRN, (e) MC-Dropout, (f) PU-CNN, (g) LU-CNN, and (h) PL-CNN.

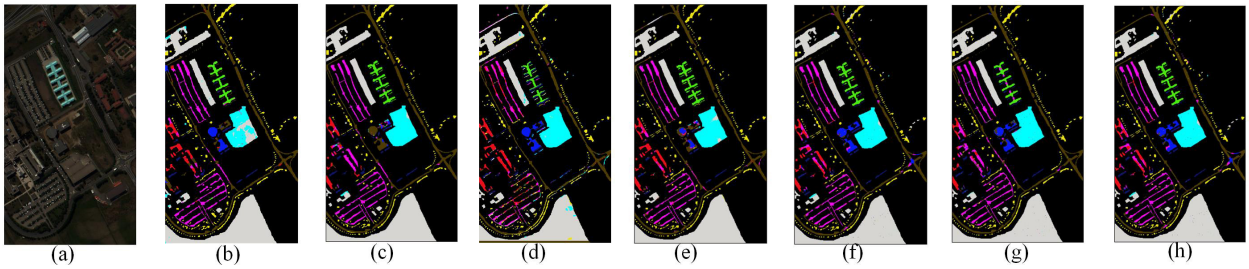


Fig. 16. Pavia. (a) False-color composite image. The classification maps using (b) EMP-SVM, (c) CNN, (d) SSRN, (e) MC-Dropout, (f) PU-CNN, (g) LU-CNN, and (h) PL-CNN.

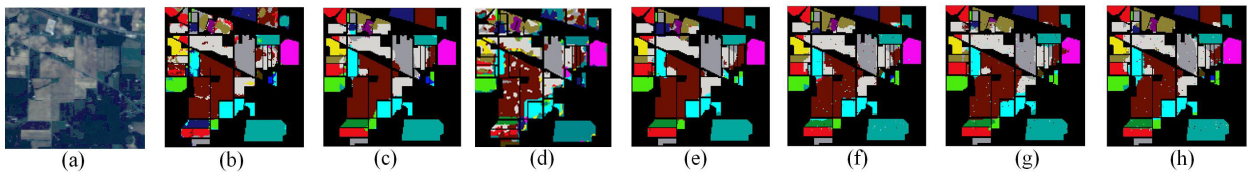


Fig. 17. Indian Pines. (a) False-color composite image. The classification maps using (b) EMP-SVM, (c) CNN, (d) SSRN, (e) MC-Dropout, (f) PU-CNN, (g) LU-CNN, and (h) PL-CNN.

the Pavia dataset. For the Salinas and Indian Pines datasets, the conclusions were similar. The above results further validate that PL-CNN could enhance the performance of the CNN-based HSI classifier.

To illustrate that all the proposed methods are more trustworthy than other state-of-the-art methods, ECE is computed as the measurement, which is shown in the last row in Tables VII–IX. The small value of ECE leads the trustworthy methods. From the results, it can be seen that the proposed methods are able to increase the confidence level of CNN-based classifiers by reducing predictive uncertainty and label uncertainty. In comparison to the classical EMP-SVM, the value of ECE reaches 31%, 15%, and 54% on the Salinas, Pavia, and Indian Pines datasets, respectively, whose confidence level is low. Compared to the deep-learning-based methods, the value of ECE of the proposed PL-CNN is the lowest, which is below 3% on all the datasets. All the above results demonstrate that the proposed methods are confident.

I. Classification Map

The classification maps of test methods are visually shown in Figs. 15–17 on the Salinas, Pavia, and Indian Pines datasets, respectively. It is obvious that the compared method EMP-SVM exhibits noisier estimations [see Figs. 15–17(b)], especially for the boundary between different classes. In addition, for the deep learning-based methods (CNN, SSRN, and MC-Dropout), there are also suffered from sparse noise [see Figs. 15–17(c)–(e)]. However, compared to the EMP-SVM, there are more correctly classified pixels. Besides, in terms of the proposed methods, [see Figs. 15–17(f)–(h)], the proposed approaches are similar to the structure of the dataset while preserving more edge information and maintaining the distribution of the dataset. More specially, for the Pavia dataset, many pixels are classified correctly in the middle, such as the class of bare soil with the blue color. However, other competitive approaches provide the map with more errors. From these figures, it can be seen that the proposed methods have better performance in comparison to other methods.

J. Running Time

The computational time of different approaches is listed in Table X. All the experiments are run on a computer with an Intel Core i7-11700 processor with 2.5 GHz, 32 GB of dual data rate (DDR4) random access memory (RAM), and an NVIDIA GeForce RTX 3070 Ti graphical processing unit (GPU).

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TABLE X
TRAINING AND TEST TIMES OF DIFFERENT TIMES ON ALL THE DATASETS

Dataset	Method	EMP-SVM	CNN	SSRN	MC-Dropout	HybridSN	**PU-CNN**	**LU-CNN**	**PL-CNN**
Salinas	Training Time (sec.)	2.27	130.64	190.77	98.35	77.57	176.56	34.96	185.35
	Test Time (sec.)	1.03	98.14	106.09	72.47	220.53	77.55	58.05	77.55
Pavia	Training Time (sec.)	3.52	41.02	43.83	34.8	75.8	95.73	21.2	110.26
	Test Time (sec.)	0.14	41.5	34.53	35.17	179.12	45.17	23.94	54.76
Indian Pines	Training Time (sec.)	1.4	32.46	168.05	31.79	395.33	85.95	34.73	92.18
	Test Time (sec.)	0.22	13.67	17.91	10.35	93.08	14.19s	6.56	13.86

Compared to the reference methods, the proposed PU-CNN and PL-CNN take a longer computational cost because the proposed PU-CNN computes the Jacobian during the training process, but it has a shorter test time, which is similar to the classical CNN. For the proposed PU-CNN, the training and test times yield considerably faster in comparison to deep learning-based methods. Specifically, the training of the proposed LU-CNN only takes 34.96, 21.20, and 34.73 seconds on the Salinas, Pavia, and Indian Pines datasets, respectively. All the proposed approaches are competitive in the classification performance and also computationally efficient on the Indian Pines dataset compared to SSRN and HybridSN.

IV. CONCLUSION

In this study, the idea of HSI trustworthy classification was investigated for the first time. Three models, i.e., PU-CNN, LU-CNN, and PL-CNN, were proposed to build trustworthy classifiers with high accuracy. First, PU-CNN explored a distance measurement scheme by creating different centroids to obtain the confidence scores, which provides an accurate confidence level on the HSI predictions of CNN. Next, LU-CNN proposed a new classifier-consistent estimator by computing loss over candidate sets, which recovered labels corresponding to different classes in each HSI spectral–spatial training sample. Finally, PL-CNN built a trustworthy system for HSI classification in the combination of prediction uncertainty and label uncertainty. The proposed methods improved the accuracy and trustworthiness at the same time.

In future work, reducing the uncertainty caused by the model structure or the training procedure is of crucial importance to establish the HSI trustworthy classifier. Other advanced techniques based on CNN can be considered for HSI classification to estimate and reduce uncertainty. For example, the ensemble is an effective technique to model uncertainty by generating distributional estimates of model uncertainty. The success of the ensemble depends on the variance reduction generated by combining predictions that are prone to several types of errors individually neural architecture distribution search and finds an appropriate distribution of different architectures. In addition, integrating the impact of uncertainty with semisupervised and active learning for acquiring new samples is also an open research direction. Activate learning with uncertainty could improve the data labeling quality by having less uncertainty than in previous studies.

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